Syllabus
Combinatorics
MATH 31003-01 - Fall 2026

Course	Combinatorics MATH 31003-01
Semester	Fall 2026
Instructor	Gennady Uraltsev
Email	uraltsev@uark.edu
Office	SCEN 421
Class meetings	Monday, Wednesday, Friday 8:35-9:25 a.m., GEAR 108
Office hours	Wednesday 4:00-5:00 p.m., SCEN 421

0.1 Course overview

See the University of Arkansas mathematics course listing. Prerequisites are MATH 26103 or MATH 28003; the pre- or corequisite is MATH 30803 or MATH 30903. You should be comfortable working with sets, functions, relations, basic proofs and mathematical arguments, algebraic manipulation, and polynomials.
Combinatorics begins with a simple question: How many ways can we arrange a collection of objects? Can a schedule, pairing, or design be made at all? What changes when two arrangements should count as the same? These questions show up in mathematics, computer science, probability, and more.
We will learn foundational combinatorial techniques and the inclusion-exclusion principle, which allow us to count efficiently. We will then learn generating functions as algebraic bookkeeping for combinatorial quantities, followed by recurrence relations that model how later states depend on previous ones. Additional topics may include modular arithmetic and applications to cryptography, coding theory, graph theory, counting arrangements with symmetries, and elementary algebraic structures.

Unit	Main capability
Enumeration and inclusion-exclusion	Define objects precisely and count them using cases, bijections, double count
Generating-function foundations	Encode sequences as formal power series and interpret algebraic operations a
Recurrence models	Select states and initial conditions, derive and solve recurrences, and compar
Further topics	To be announced.

0.1.1 Learning outcomes

By the end of the course, you should be able to:

1. Use basic counting techniques, including permutations, combinations with repetition, and careful accounting for overcounting.
2. Analyze a discrete model by identifying its objects, states, parameters, and constraints.
3. Choose and switch representations to reveal a model's useful structure.
4. Explain a combinatorial argument clearly: describe the model, choose a representation, and apply the right known techniques.
5. Describe a recursive model and analyze its behavior by identifying state information, a recurrence, and initial conditions, and by finding a closed form when possible.
6. Use generating functions as formal algebraic models and extract the information they encode.
7. Work with selected advanced discrete systems, which may include symmetry, graphs, algorithms, and complexity analysis.

0.1.2 Materials

Our lecture notes are the main guide through the course. The principal reference is Keller and Trotter, *Applied Combinatorics*: web edition and PDF edition. Additional references are listed on the resources page; you do not need to purchase any of the listed open materials.

0.2 Course organization

In class, I will introduce new topics and talk through their applications with you. After most meetings, online questions provide a quick review of basic skills; they are graded automatically, and most allow multiple attempts. I will assign hand-in homework weekly. We will have three midterms and a final-period exam.

0.2.1 Attendance and participation

Attendance is not graded, and there is no absence penalty. If you need to miss class, use the posted notes, online questions, homework, and other resources to keep up independently. Please keep track of course material, announcements, deadlines, and assessments. If your work suggests that your current study routine is not serving you, I may reach out with ideas such as regular attendance or office hours.

Please plan to be present for scheduled exams. I will not be able to offer make- up exams except for excused emergencies.

0.2.2 Academic integrity

We all share responsibility for creating a course environment built on honest work and learning. Follow the University of Arkansas Academic Integrity Policy. Specific expectations for collaboration and artificial intelligence appear on the assignments page.

0.3 Assessment

Component	Weight
Written hand-in homework	20%
Online questions	15%
Exam 1	16%
Exam 2	16%
Exam 3	16%
Exam 4	16%
Course check-in completion	1%
Total	100%

Course check-ins are brief questions I create to learn what is working for you in the course organization, pacing, and resources.

0.3.1 Dropped assignments

- The worst two hand-in homework assignments are dropped.
- The worst four online-question assignments are dropped. One extra online question is dropped if you complete all check-in work.

0.3.2 Weekly hand-in homework

A typical assignment has several problems and may involve computation, modeling, proofs, or other mathematical arguments. Submit it online as a PDF; a handwritten, scanned PDF is welcome. Make sure it is legible, labels each problem, explains the method and final answer, and uses complete mathematical statements. Follow the deadline listed on the assignment page.

If you need more time, talk to me in advance of the deadline so we can work out an extension. Unforeseen emergencies are handled case by case.

0.3.3 Online questions

Online questions normally follow a lecture and are due by the start of the next class meeting. They focus on comfort with key vocabulary and formulas.

0.3.4 Unit exams

There are four unit exams: three in-class exams and one during the first part of the final-exam period. Every exam problem will be parallel to an assigned homework problem in its required method and main solution steps, though the parameters, objects, or wording may change.

Exam	Date or period	Scope
Exam 1	Friday, September 18	To be announced
Exam 2	Friday, October 16	To be announced
Exam 3	Friday, November 13	To be announced
Exam 4	First hour of the Registrar-assigned final exam	To be announced
Mastery reassessment	Second part of the same final period	One earlier midterm unit selected by the s

During the second part of the final exam period, you may choose to reassess your mastery of one of exam units 1–3. The reassessment covers the same unit and methods. If the reassessment score is higher, it replaces the earlier score; if it is lower, the earlier score remains.

For each exam, you may use one standard letter- or A4-size page of handwritten notes that you prepared yourself. If you need an alternate or accommodated testing arrangement, contact me and the appropriate University office at least two weeks in advance.

0.3.5 Letter grades

The following tentative distribution is a reference; I will let you know if it needs to change during the course.

Course percentage	Letter grade
88–100	A
76–87.99	B
64–75.99	C
51–63.99	D
Below 51	F

0.4 Communication, records, and course changes

Email is the best way to reach me with an individual question. I will post course-wide announcements, grades, and updates in Blackboard, so check your University email and Blackboard regularly. If you notice a possible grading or recording error, let me know promptly, within one week of the posted score.

I may revise the syllabus when needed. If a change materially affects grading, exams, deadlines, or course delivery, I will announce it in writing. The current version of this website is our course reference.

0.5 Other information

0.5.1 Accessibility and accommodations

If you have approved accommodations, request your accommodation letter and contact me early so we can make arrangements in time. If you think you may need accommodations but have not set them up, the Center for Educational Access can help. Temporary conditions can affect access too, so reach out promptly when they arise.

0.5.2 Religious observances, University activities, and other excusable conflicts

Email me as early as you can about religious observances, required University activities, military service, jury duty, documented illness, a family crisis, or another excusable conflict. For an exam conflict you can anticipate, letting me know at least one week ahead is especially helpful. We will follow applicable University policies and deadlines.
If you have three or more final exams on one day, you may be able to reschedule one under Academic Policy 1500.20. If it involves this course, request it before the Registrar's withdrawal deadline.

0.5.3 Inclement weather and emergencies

We will follow University closure and inclement-weather decisions. If a closure prevents an exam, I will announce a new date through Blackboard and the course website. Please do not travel when conditions are unsafe.

- University emergency information
- RazALERT emergency notifications
- Academic Policy 1858.10: inclement weather

0.5.4 Student support

- Mathematics Resource and Teaching Center
- Student Success
- Counseling and Psychological Services
- U of A Cares

Additional links are collected on the resources page.